

Group 25: RSA Televisions Write-Up

Considering that at 25 million revenue with \$700 price per TV results in 35,700 units per year and penalties for oversupply is higher than under supply we can approximately calculate the best course of action for the COO.

Given the EOQ formula and considering:

Annual demand: 35,700 units

Monthly demand: $35,700/12 \text{ months} = 2975$

Weekly demand: $35,700/52 = 687$

Holding cost per unit: \$490 (\$700 * 30% profit) multiplied by an unknown cost that has increased by 15% spike.

This gets us a formula:

$$f(x) = \sqrt{\frac{2(35700)(2975)}{(490)(x)(1.15)}}$$

Short-term:

With annual demand running at 35 700 TVs (~ 2 975 a month, 687 a week) and the 3-week factory-yard backlog plus 7-to-10-week ocean leg stretching total lead time to roughly 12 weeks, the reorder trigger sits around 9 000 units (687×12 , rounded to the nearest pallet lot). Plugging the same demand into the EOQ formula and using a mid-range 25 % carrying-cost rate (already inflated by the 15 % jump in storage) rounds to one 3 000-unit sailing every 4-5 weeks. That cadence keeps exactly two shipments on the water at any time, covers in-transit demand, and, because the \$30 k penalty for stock-outs outweighs the \$25 k hit for excess, we should lean very slightly toward having a buffer without tying up more working capital than necessary.

Long-term:

When storage falls back to 1 week and the voyage itself to about 3 weeks, total lead time collapses to ~ 4 weeks; the reorder point, therefore, drops to only ~ 2 700 – 3 000 TVs. At the same time demand is forecast to rebound toward ~ 57 000 TVs a year as retailers reopen and consumer spending normalises. Re-running EOQ with that higher volume and a lower per-unit carrying cost pushes the economic lot to ~ 4 000 units which should now be ordered every 3–4 weeks. Regular quarterly reviews of demand, carrying cost, and lead time will tell the COO when to nudge that lot size to 5 000 or shorten the cadence further, keeping total landed cost down while still hitting Womart's 90 % service-level requirement.

What impact do you believe labor shortages and storage issues will have?

Even if production returns at full scale, RSA's capacity to process, package, and ship televisions would be severely limited due to labor shortages at distribution centers, which have decreased capacity by 25%. There is currently a risk of a backlog once manufacturing ramps up because each center can only handle 7,500 units per month. Considering storage, the 15 % jump in China yard fees, combined with increased storage times of 3 weeks, roughly triples holding cost on finished TVs. That extra fee per set directly reduces the optimal shipment size. If the bottleneck breaks before yard rates fall, RSA will be paying more to hold the larger buffers that higher output demands.

What should be done about reduced sales?

To reverse declining sales and regain retailer confidence, RSA must first stabilize its supply chain. Production and delivery consistency are essential prerequisites for revenue recovery. This requires rapid execution of short-term operational fixes such as restoring throughput, managing logistics delays, and closing fulfillment gaps while preserving long-term strategic partnerships, particularly with Walmart. In addition, a lot of the sales numbers are from covid times which were artificially inflated by panic-driven demand and supply constraints, making them unreliable in predicting long-term future gains.

What strategies should be put in place for distribution delays, shipping delays, manufacturing delays etc. ?

RSA needs to stay flexible and better coordinated in the short term. Weekly check-ins between production, logistics, and sales teams can ensure everyone is working with the most current information, helping us adjust shipments quickly when delays happen or inventory needs shift. For shipping, partnering with multiple freight companies gives us backup options, and we should prioritize sending out our most in-demand models first. On the manufacturing side, we can temporarily outsource a portion of production or rotate which models we focus on to make the most of our limited capacity. While we may not be able to respond to every issue right away, focusing on high-priority products and clear communication can help reduce the impact of delays as we work toward long-term improvements like expanded production and better forecasting.

2. Given the current trajectory with exports, leadtimes etc., what strategies should RSA implement to ensure inventory is available?

To ensure inventory availability, RSA should utilize demand forecasts and maintain buffer stock at distribution centers, particularly for high-demand products, to allow both productivity and flexibility. They should do this by using a detailed inventory to keep track of the inventory at all times. With warehouse rates down, securing short-term storage will overall reduce delivery delays without committing to long-term costs. Additionally, improving turnaround time, like the speed at which inventory moves from storage to delivery, will help prevent stock from sitting idle and ensure that retailers receive products faster, even in the face of shipping or manufacturing disruptions.